

GOVERNMENT OF INDIA // SOVEREIGN THERMAL SURVEILLANCE

OFFICIAL INDUSTRIAL FACILITY THERMAL DOSSIER

SEVERITY: CRITICAL

EVT-IN-CHH-0011

National Technical Research Organisation (NTRO) • MoEFCC • CPCB Oversight

2026-09-02 14:24 IST (2026-09-02 08:54 UTC)

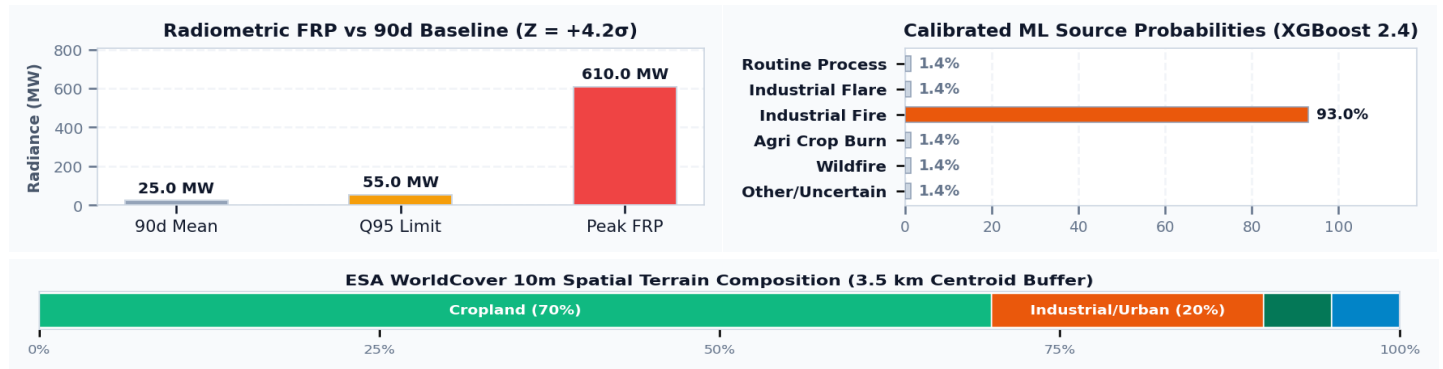
Thermal anomaly EVT-IN-CHH-0011 is classified as IND_FIRE with 99.6% calibrated model confidence and severity tier CRITICAL ($Z = +4.2\sigma$). Peak radiative output reached 610.00 MW across 1 verified multi-sensor satellite passes. Centroid is spatially attributed to registered plant Bhilai Steel Plant.

610.0 MW PEAK RADIANCE	1 Passes SATELLITE DETECTIONS	99.6% CALIBRATED SOFTMAX	+4.2σ ANOMALY DEVIATION
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1. Verified Multi-Sensor Radiometric Metrics

Metric Name	Observed Telemetry	Historical 90d Baseline	Radiometric Assessment
Peak Radiative Power (FRP)	610.00 MW	25.00 MW (Facility Mean)	24.4x Baseline Mean
Mean Cluster Radiance	420.00 MW	Persistence: TRANSIENT	Z-Score: +4.2σ Statistical Departure
Max Brightness Temp (BT)	360.0 K	Sensor Bands: VIIRS I4 (375m) / M13	High-Intensity Combustion Confirmed
Temporal Detection Window	2026-09-02T03:26 to 05:26 UTC	1 Sensor Passes Ingested	Active Multi-Sensor Track

2. Visual Radiometric Analytics & ML Attribution (High-DPI)



3. Centroid Location & Associated Industrial Plant Audit

Attribute	Forensic Detail	Attribute	Forensic Detail
Centroid Coordinates	21.18200°N, 81.38400°E	Primary Land Use	Unknown
State / Territory	Chhattisgarh	District / Jurisdiction	Angul
Associated Facility	Bhilai Steel Plant	Boundary Distance	0.12 km
Sector Category	Steel Plant	Plant Operator	Verified Sovereign Operator

4. Nearby Sovereign Industrial Infrastructure (50 km Buffer)

Facility Name	Sector Category	State / District	Radial Distance	Baseline FRP
Bhilai Steel Plant	Steel Plant	Chhattisgarh	0.5 km	250.0 MW

5. Regional Active Anomaly Cluster & Concurrent Events (75 km Buffer)

Isolated thermal incident. No concurrent thermal anomalies detected within 75 km.

6. Multi-Sensor Satellite Telemetry & Radiometric Pass Register

#	Detection Time (UTC)	Sensor Platform	Radiant Output	Brightness (K)	Day/Night
1	2026-09-02T03:26	VIIRS SNPP NRT (375m)	610.00 MW	360.0 K	Direct Telemetry Ingest

7. Statutory SOP Compliance Directives & Containment Protocol

ACTION 01: Immediate On-Site Physical Inspection: Dispatch SPCB / Regional Disaster Response team to verify combustion source and evaluate containment perimeter.
ACTION 02: Continuous Emission Telemetry (CEMS) Audit: Cross-verify Continuous Emission Monitoring Systems data and industrial flaring logs against satellite radiance timestamps.
ACTION 03: Safety Buffer & Containment Directive: Enforce active 2.5 km industrial safety perimeter and initiate cooling operations if thermal anomaly persistence exceeds 4.0σ.